

**A Vision-Based Intelligent Overtaking Decision
Support Framework for Improving Road Safety on
Two-Lane Roads**

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1. Research Gap

Previous studies have investigated overtaking safety using predictive models, computer vision, V2V communication, driver behavior analysis, and autonomous vehicle systems. Fadhil and Al-Bayatti studied predictive overtaking assistance, while Yang et al. investigated camera-based vehicle distance estimation on two-lane roads. Other studies have examined Time-to-Collision (TTC), safe time gaps, overtaking behavior, driver guidance, and autonomous overtaking. These studies show that different parts of the overtaking problem have been addressed, but there is still an opportunity to investigate an integrated software-based framework that combines vehicle perception and predictive safety analysis to provide a clear overtaking recommendation to human drivers.

Previous studies have investigated overtaking safety from several perspectives, including driver behavior, predictive models, vehicle-to-vehicle communication, computer vision, autonomous vehicle trajectory planning, and human-machine interfaces. Fadhil and Al-Bayatti proposed a Driver Overtaking Assistance System using predictive models and considered driver, vehicle, road, and environmental factors. Yang et al. investigated monocular vision and YOLOv5s for measuring the distance of forward and opposing vehicles on two-way two-lane roads. Other research has investigated how overtaking advice can be presented to drivers through an HMI.

Research has also examined the safety of overtaking using Time-to-Collision (TTC), accepted time gaps, and passing duration. Mwesiga et al. used TTC as a surrogate safety measure for assessing the risk of passing maneuvers on two-lane rural highways. Farah studied the conditions under which drivers abort an overtaking maneuver and found that factors such as the accepted gap, vehicle speeds, vehicle type, road curvature, and driver characteristics influence the outcome.

More recent research has focused on autonomous overtaking, including trajectory planning and personalized decision-making. Vigne et al. proposed an automated overtaking framework for two-lane two-way rural roads, while Palatti et al. investigated safe overtaking and aborting maneuvers for autonomous vehicles. Research on overtaking datasets has also identified limitations in available real-world overtaking data.

Identified Research Gap

Existing studies have investigated predictive overtaking assistance, vehicle perception, safety analysis, and autonomous overtaking, there is still an opportunity to investigate an integrated software-based, vision-assisted decision-support framework that combines surrounding-vehicle perception with predictive overtaking safety analysis and provides a simple recommendation to a human driver before the overtaking maneuver begins, with consideration of mixed-traffic two-lane environments.

2. Problem Statement

Existing research has investigated overtaking assistance using predictive models, computer vision, and autonomous-driving technologies, these studies address different aspects of the overtaking problem, such as vehicle perception, trajectory planning, risk estimation, or driver guidance. There is still a need to investigate how these capabilities can be integrated into a software-based decision-support framework that evaluates surrounding traffic conditions and provides a clear overtaking safety recommendation before the maneuver begins, particularly for mixed-traffic two-lane environments.

3. Aim

To design and investigate a software-based intelligent decision-support framework that uses vision-based traffic perception and predictive analysis to evaluate overtaking safety on two-lane roads.

4. Objectives

- To study the factors that influence safe and unsafe overtaking decisions on two-lane roads.
- To review existing overtaking assistance, vehicle perception, and road safety approaches.
- To investigate computer-vision-based methods for obtaining relevant traffic information from the road environment.
- To investigate prediction and mathematical approaches for estimating overtaking safety using traffic parameters such as speed, distance, and available time gaps.
- To design a software-based decision-support framework that integrates traffic perception and predictive overtaking analysis.
- To investigate an appropriate method for presenting the predicted overtaking recommendation to the driver.

5. References

Overtaking assistance

- Fadhil & Al-Bayatti (2022) — predictive overtaking assistance.
- Fadhil (2023) — Bayesian Networks for the Driver Overtaking Assistance System on Two-Lane Roads. The 2022 paper's indexing page identifies this follow-up publication and DOI 10.3311/PPtr.18459.
- Perumal et al. (2021) — crash avoidance and overtaking advice review.
- Vigne et al. (2024) — automated overtaking on two-lane rural roads.
- Palatti et al. (2021) — safe abortable overtaking.

Computer vision / vehicle perception

- Yang et al. (2023) — monocular vision and YOLOv5s distance measurement.
- Research on vehicle detection/tracking and road/lane perception.
- Research on monocular depth/distance estimation.
- Research on deep-learning vehicle detection.
- Research on vehicle tracking.

Overtaking behavior and safety

- Farah (2016) — aborted overtaking.
- Mwesiga, Farah & Koutsopoulos (2016) — passing-zone risk and TTC.
- Research on overtaking gap acceptance.
- Research on overtaking behavior.
- Research on two-lane road safety.

Human-machine interaction

- Fu et al. (2023) — overtaking assistance HMI.
- Research on driver warnings.
- Research on ADAS user interfaces.

Autonomous overtaking / prediction

- Palatti et al. (2021) — overtaking planning.
- Pinto, Dutra & Fonseca (2023) — overtaking datasets and knowledge.

